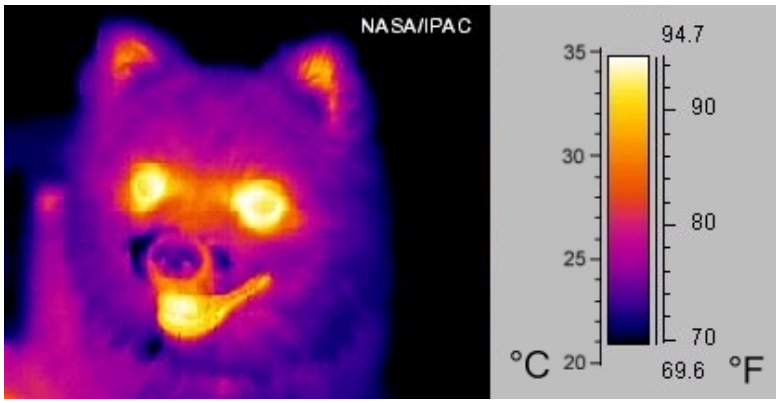
Infrared



Picture of dog in mid-infrared

Infrared (**IR**) radiation is a type of electromagnetic radiation (a wave with electricity) . The wave is longer than light which humans can see and shorter than microwaves. The word *infrared* means *below red*. It comes from the Latin word *infra* (meaning *below*) and the English word *red*. (Infrared light has a frequency *below* the frequency of red light.) Red light has the longest wavelength that human eyes can see. Infrared waves cannot be seen by the eye. The infrared wave is between 750 nm and 1 mm. People sense infrared as heat. Most remote controls use infrared to send the control signals.

Telecommunications

Before Bluetooth was invented, some computers, personal digital assistants, and mobile phones were using the infrared technology to send files to other devices. Bluetooth replaced infrared in the early 2000's. Infrared is limited by the need to have both devices in a "line of sight" with each other.[1]

Infrared lasers are used to provide the light for optical fiber communications systems. Infrared light with a wavelength around 1,330 nm (least dispersion) or 1,550 nm (best transmission) are the best choices for standard silica fibers.

Spectroscopy

Infrared vibrational spectroscopy (IR spectroscopy) can be used to identify molecules by listing their chemical bonds. An IR spectrometer is a machine that shines infrared light from a set range of frequencies on a sample. It measures which exact light frequencies are absorbed by the sample.

Each chemical bond in a molecule vibrates at a frequency which is characteristic of that bond. A group of atoms in a molecule may have multiple modes of oscillation caused by the stretching and bending motions of the group as a whole. If an oscillation leads to a change in dipole in the molecule, then it will absorb a photon which has the same frequency. The vibrational frequencies of most molecules correspond to the frequencies of infrared light. Typically, the technique is used to study organic compounds using light radiation from 4000–400 cm^{-1}, the mid-infrared. The spectrometer records all the frequencies of absorption in a sample. This can be used to gain information about the sample composition in terms of chemical groups present and also its purity (for example a wet sample will show a broad O-H absorption around 3200 cm^{-1}).

References

Article Sources and Contributors

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